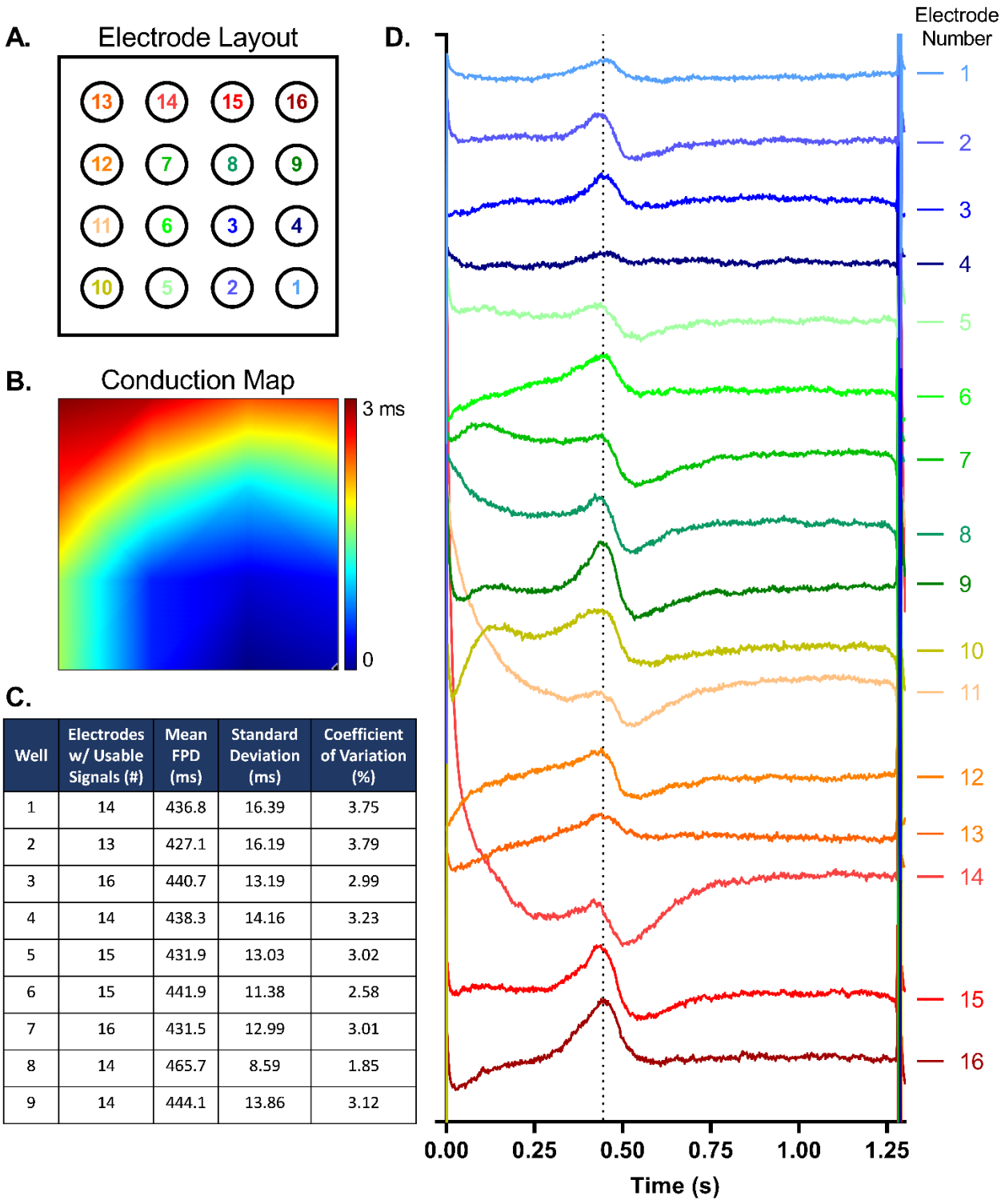
Intrawell FPD Variability



C.

Well	Electrodes w/ Usable Signals (#)	Mean FPD (ms)	Standard Deviation (ms)	Coefficient of Variation (%)
1	14	436.8	16.39	3.75
2	13	427.1	16.19	3.79
3	16	440.7	13.19	2.99
4	14	438.3	14.16	3.23
5	15	431.9	13.03	3.02
6	15	441.9	11.38	2.58
7	16	431.5	12.99	3.01
8	14	465.7	8.59	1.85
9	14	444.1	13.86	3.12

Supplemental Figure 1. iCell Cardiomyocytes² intrawell field potential duration (FPD) variability. A) Schematic indicating electrode layout within each well. **B)** Representative conduction map demonstrating uniform conduction across a well. **C)** Measured FPD values were similar across all of the electrodes within a well, as demonstrated by the low coefficient of variation. **D)** FPD traces from each electrode within well #6 of the table shown in panel C. Electrode numbers are indicated on the right of the figure and correspond to the electrode map shown in panel A.

Supplemental Table 1: Spontaneous iCell Cardiomyocytes² electrophysiology values across 20 minutes of equilibration in a microelectrode array system. BPM: beats per minute, FPD: field potential duration, stdev: standard deviation. Ordinary one-way ANOVA with multiple comparisons to minute 20 using Holm-Sidak correction; significant p-values are bolded. Cohen effect size indicated (absolute value), relative to minute 20.

Minute	Beat Rate (BPM)				FPD (ms)				Spike Amplitude (mV)				Conduction Velocity (mm/ms)			
	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value
2	40.18 ± 2.71	**<0.0001**	1.979	120	385.8 ± 19.8	**<0.0001**	1.538	108	1.108 ± 0.298	0.9415	0.188	117	0.252 ± 0.085	0.9320	0.185	107
3	41.74 ± 3.40	**<0.0001**	1.564	120	379.0 ± 19.6	**<0.0001**	1.192	112	1.070 ± 0.297	>0.9999	0.065	118	0.260 ± 0.097	0.9983	0.092	109
4	43.20 ± 4.06	**<0.0001**	1.199	120	373.5 ± 21.0	**<0.0001**	0.873	115	1.053 ± 0.295	>0.9999	0.009	119	0.260 ± 0.092	0.9983	0.095	110
5	44.45 ± 4.46	**<0.0001**	0.923	120	367.9 ± 21.3	**<0.0001**	0.587	117	1.053 ± 0.291	>0.9999	0.009	119	0.261 ± 0.092	0.9983	0.084	111
6	45.34 ± 4.75	**<0.0001**	0.735	120	364.9 ± 21.0	**0.0052**	0.442	118	1.029 ± 0.288	>0.9999	0.069	119	0.267 ± 0.094	0.9983	0.021	113
7	45.97 ± 4.91	**<0.0001**	0.609	120	363.4 ± 20.7	**0.0413**	0.369	119	1.030 ± 0.285	>0.9999	0.066	119	0.267 ± 0.087	0.9983	0.022	116
8	46.37 ± 5.02	**0.0001**	0.531	120	361.7 ± 19.0	0.2473	0.296	116	1.034 ± 0.296	>0.9999	0.052	119	0.262 ± 0.084	0.9983	0.077	114
9	46.72 ± 5.07	**0.0010**	0.465	120	360.5 ± 19.5	0.4731	0.230	117	1.027 ± 0.297	>0.9999	0.075	119	0.261 ± 0.085	0.9983	0.087	113
10	46.94 ± 5.19	**0.0036**	0.421	120	360.8 ± 18.8	0.4190	0.250	119	1.034 ± 0.299	>0.9999	0.052	119	0.258 ± 0.085	0.9983	0.120	111
11	47.12 ± 5.26	**0.0087**	0.386	120	361.7 ± 19.6	0.2473	0.291	119	1.038 ± 0.308	>0.9999	0.038	119	0.258 ± 0.088	0.9983	0.118	111
12	47.37 ± 5.27	**0.0278**	0.341	120	360.9 ± 18.3	0.4059	0.259	119	1.050 ± 0.304	>0.9999	0.000	119	0.259 ± 0.086	0.9983	0.108	113
13	47.55 ± 5.34	0.0551	0.307	120	359.9 ± 18.4	0.5982	0.204	119	1.042 ± 0.306	>0.9999	0.026	119	0.264 ± 0.089	0.9983	0.053	114
14	47.82 ± 5.34	0.1452	0.259	120	359.2 ± 18.2	0.7511	0.168	119	1.046 ± 0.290	>0.9999	0.013	118	0.263 ± 0.089	0.9983	0.064	115
15	48.01 ± 5.47	0.2349	0.223	120	359.1 ± 18.0	0.7511	0.163	119	1.050 ± 0.316	>0.9999	0.000	119	0.259 ± 0.085	0.9983	0.109	113
16	48.29 ± 5.53	0.4309	0.173	120	358.8 ± 17.6	0.7511	0.148	118	1.047 ± 0.321	>0.9999	0.009	118	0.258 ± 0.089	0.9983	0.118	110
17	48.46 ± 5.62	0.5099	0.142	120	358.6 ± 17.7	0.7511	0.137	117	1.035 ± 0.300	>0.9999	0.048	118	0.265 ± 0.094	0.9983	0.042	110
18	48.75 ± 5.74	0.6569	0.091	120	356.7 ± 17.9	0.9549	0.033	118	1.055 ± 0.333	>0.9999	0.015	119	0.257 ± 0.087	0.9979	0.130	108
19	49.01 ± 5.82	0.6868	0.046	120	356.1 ± 17.8	0.9978	0.000	118	1.046 ± 0.316	>0.9999	0.013	118	0.261 ± 0.090	0.9983	0.085	108
20	49.28 ± 5.91	-	-	120	356.1 ± 18.8	-	-	118	1.050 ± 0.319	-	-	118	0.269 ± 0.098	-	-	111

Supplemental Table 2: Spontaneous iCell Cardiomyocytes² electrophysiology values across 14 days of culture. BPM: beats per minute, FPD: field potential duration, stdev: standard deviation. Ordinary one-way ANOVA with multiple comparisons to day 7 using Holm-Sidak correction; significant p-values are bolded. Cohen effect size indicated (absolute value), relative to day 7 measurements.

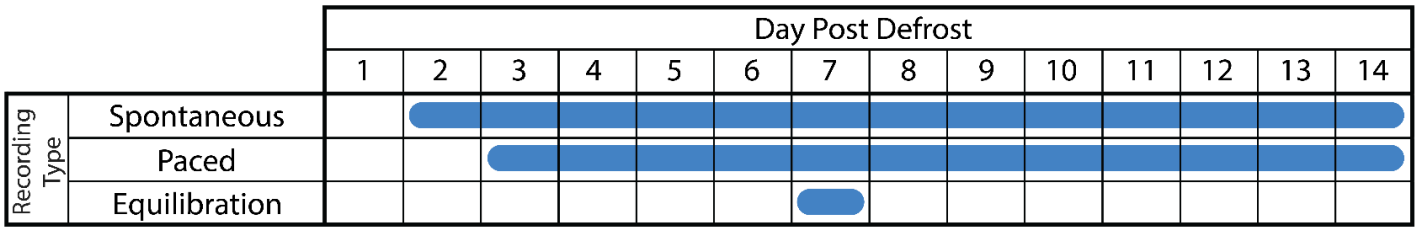
Day	Beat Rate (BPM)				FPD (ms)				Spike Amplitude (mV)				Conduction Velocity (mm/ms)			
	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value
2	53.80 ± 5.12	**<0.0001**	1.146	148	288.9 ± 15.7	**<0.0001**	5.380	141	0.301 ± 0.238	**<0.0001**	3.806	167	0.328 ± 0.095	**<0.0001**	0.906	63
3	49.91 ± 8.29	**<0.0001**	0.458	264	352.6 ± 27.3	**<0.0001**	2.886	248	0.860 ± 0.438	**<0.0001**	2.559	257	0.291 ± 0.088	**<0.0001**	0.465	224
4	48.32 ± 5.59	**0.0001**	0.309	363	372.1 ± 27.6	**<0.0001**	2.304	351	1.162 ± 0.349	**<0.0001**	2.174	358	0.262 ± 0.089	0.6867	0.079	344
5	49.48 ± 6.10	**<0.0001**	0.464	138	412.8 ± 24.9	**<0.0001**	1.145	134	1.701 ± 0.419	**<0.0001**	1.220	135	0.207 ± 0.046	**<0.0001**	0.917	133
6	46.30 ± 4.20	0.9112	0.013	228	435.1 ± 23.7	**<0.0001**	0.476	223	2.023 ± 0.510	**<0.0001**	0.661	232	0.252 ± 0.060	0.8214	0.067	203
7	46.22 ± 7.83	-	-	268	450.6 ± 39.5	-	-	255	2.452 ± 0.763	-	-	267	0.256 ± 0.060	-	-	240
8	45.89 ± 5.25	0.9112	0.050	68	465.2 ± 24.9	**0.0012**	0.442	62	2.707 ± 0.763	**0.0026**	0.334	68	0.258 ± 0.058	0.8528	0.034	62
9	42.98 ± 5.11	**0.0005**	0.490	66	499.7 ± 32.5	**<0.0001**	1.357	61	2.637 ± 0.621	**0.0192**	0.266	67	0.286 ± 0.070	**0.0265**	0.460	63
10	39.41 ± 6.80	**<0.0001**	0.929	75	505.9 ± 59.5	**<0.0001**	1.095	82	2.691 ± 0.647	**0.0026**	0.338	86	0.353 ± 0.116	**<0.0001**	1.050	74
11	39.04 ± 4.76	**<0.0001**	1.108	68	547.9 ± 37.0	**<0.0001**	2.542	65	2.987 ± 0.729	**<0.0001**	0.717	68	0.329 ± 0.075	**<0.0001**	1.075	61
12	39.64 ± 3.26	**<0.0001**	1.097	34	566.4 ± 24.3	**<0.0001**	3.531	33	3.303 ± 0.723	**<0.0001**	1.145	34	0.298 ± 0.084	**0.0236**	0.575	32
13	40.07 ± 5.96	**<0.0001**	0.884	68	550.0 ± 49.0	**<0.0001**	2.233	64	2.838 ± 1.179	**<0.0001**	0.389	72	0.347 ± 0.117	**<0.0001**	0.979	60
14	41.12 ± 4.48	**<0.0001**	0.800	40	545.9 ± 24.2	**<0.0001**	2.909	36	3.611 ± 0.774	**<0.0001**	1.509	39	0.296 ± 0.055	**0.0236**	0.695	37

Supplemental Table 3: Paced (1.5 Hz) iCell Cardiomyocytes^2 electrophysiology values across 14 days of culture. FPD: field potential duration, stdev: standard deviation. Ordinary one-way ANOVA with multiple comparisons to day 7 using Holm-Sidak correction; significant p-values are bolded. Cohen effect size indicated (absolute value), relative to day 7 measurements.

Day	FPD (ms)				Spike Amplitude (mV)				Conduction Velocity (mm/ms)			
	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value
3	275.5 ± 13.8	**<0.0001**	3.278	180	3.819 ± 1.109	**<0.0001**	1.364	216	0.239 ± 0.024	0.1950	0.255	204
4	282.1 ± 18.5	**<0.0001**	2.371	300	4.150 ± 0.969	**<0.0001**	1.149	313	0.225 ± 0.041	**0.0144**	0.241	318
5	311.5 ± 13.5	**<0.0001**	0.686	132	4.269 ± 0.916	**<0.0001**	1.061	124	0.204 ± 0.018	**<0.0001**	1.404	134
6	316.6 ± 13.8	**0.0268**	0.310	164	5.012 ± 0.898	**0.0023**	0.387	142	0.220 ± 0.021	**0.0006**	0.590	137
7	320.9 ± 13.9	-	-	151	5.434 ± 1.254	-	-	145	0.233 ± 0.023	-	-	143
8	323.6 ± 14.6	0.2339	0.189	62	5.633 ± 1.092	0.1983	0.169	64	0.226 ± 0.029	0.2082	0.267	62
9	337.6 ± 14.3	**<0.0001**	1.184	62	5.693 ± 0.829	0.1760	0.244	65	0.230 ± 0.029	0.4402	0.115	63
10	333.7 ± 13.8	**<0.0001**	0.924	55	5.940 ± 1.005	**0.0038**	0.445	62	0.256 ± 0.040	**<0.0001**	0.705	60
11	332.6 ± 15.8	**<0.0001**	0.786	43	6.596 ± 0.925	**<0.0001**	1.055	41	0.254 ± 0.022	**0.0002**	0.933	42
12	344.4 ± 14.1	**<0.0001**	1.679	32	6.548 ± 0.944	**<0.0001**	1.004	34	0.265 ± 0.014	**<0.0001**	1.681	33
13	333.9 ± 16.5	**<0.0001**	0.852	46	6.690 ± 1.378	**<0.0001**	0.953	48	0.280 ± 0.021	**<0.0001**	2.134	40
14	336.0 ± 14.0	**<0.0001**	1.082	38	6.871 ± 0.899	**<0.0001**	1.317	38	0.275 ± 0.017	**<0.0001**	2.077	35

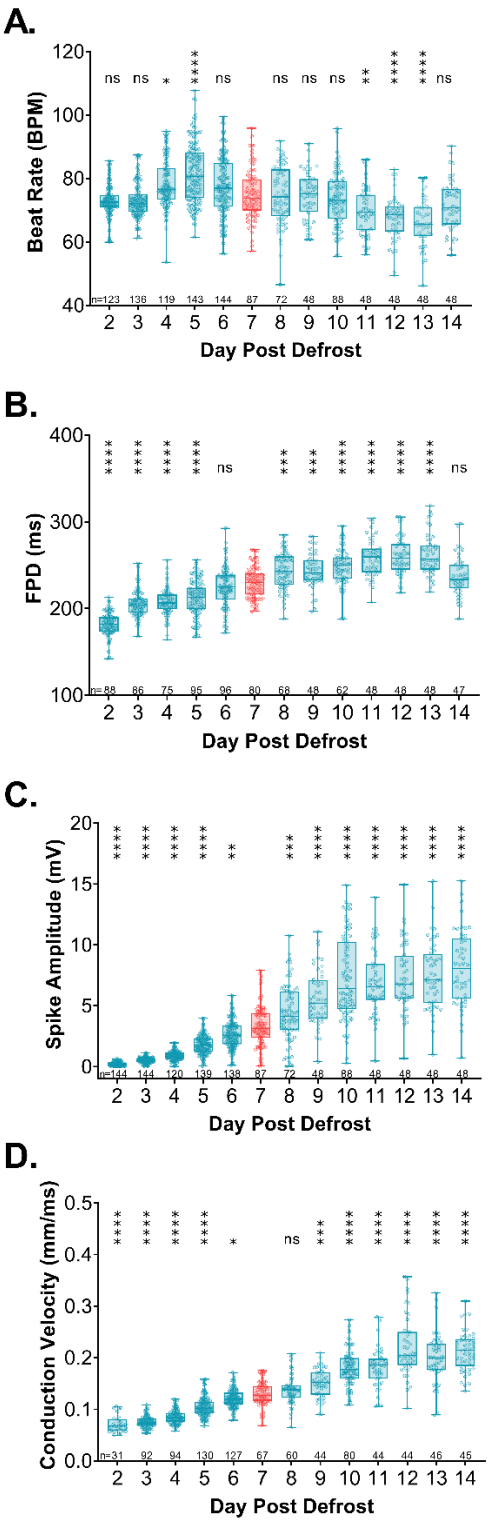
Supplemental Table 4: Paced (1.5 Hz) iCell Cardiomyocytes² action potential duration measurements from day 4 to 10 of culture. APD30: action potential duration at 30% repolarization, APD50: action potential duration at 50% repolarization, APD90: action potential duration at 90% repolarization, stdev: standard deviation. Ordinary one-way ANOVA with multiple comparisons to day 7 using Holm-Sidak correction; significant p-values are bolded. Cohen effect size indicated (absolute value), relative to day 7 measurements.

Day	APD30 (ms)				APD50 (ms)				APD90 (ms)			
	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value
4	171.7 ± 19.4	0.6239	0.509	15	247.6 ± 16.4	**0.0213**	1.143	15	319.6 ± 15.3	**<0.0001**	2.844	15
5	169.5 ± 21.0	0.6239	0.598	10	257.3 ± 6.6	0.4237	0.719	9	336.7 ± 11.7	**0.0004**	1.866	10
6	169.5 ± 13.8	0.6239	0.720	9	253.0 ± 18.6	0.2404	0.753	10	352.1 ± 19.6	0.2333	0.433	10
7	181.5 ± 19.1	-	-	17	266.0 ± 15.8	-	-	17	359.2 ± 12.4	-	-	17
8	182.2 ± 20.6	0.9308	0.035	27	269.5 ± 16.5	0.5220	0.217	27	365.9 ± 12.3	0.2333	0.543	27
9	184.5 ± 32.6	0.9308	0.112	10	279.1 ± 25.6	0.2404	0.616	10	389.7 ± 13.2	**<0.0001**	2.382	10
10	203.7 ± 26.0	**0.0260**	0.973	17	298.4 ± 20.5	**<0.0001**	1.770	17	399.2 ± 15.5	**<0.0001**	2.850	17



Supplemental Figure 2. Timeline of Cor.4U microelectrode array (MEA) recordings.

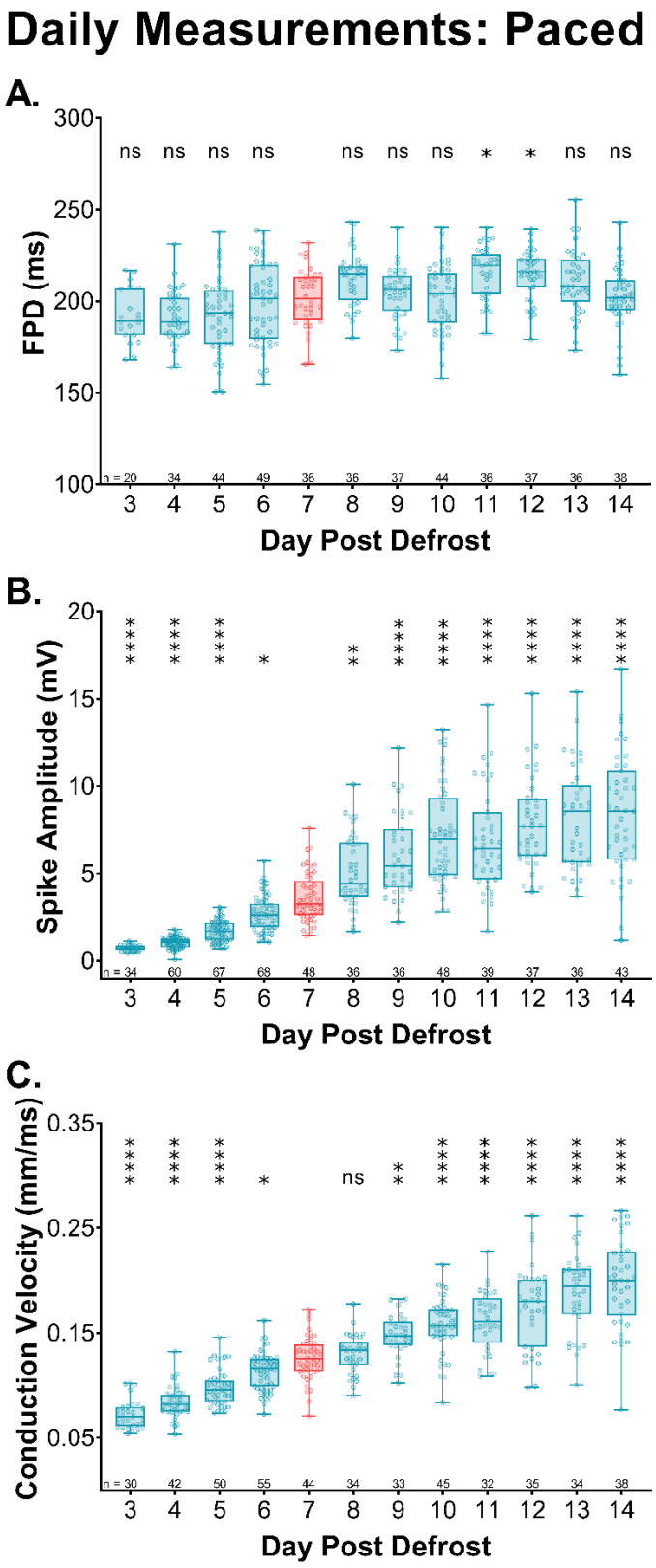
Daily Measurements: Spontaneous



Supplemental Figure 3. Daily changes in Cor.4U cardiomyocyte electrophysiology measurements (spontaneously beating). During two weeks of cell culture, **A)** spontaneous beating rate changed in a sinusoidal manner, **B)** field potential duration (FPD) lengthened, and both **C)** depolarization spike amplitude and **D)** conduction velocity increased. Daily measurements are shown as a box-and-whisker plot (all values shown, min to max), with statistical comparison versus day 7 values (shown in red). ANOVA with Holm-Sidak correction for multiple comparisons. *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. Number of replicates indicated.

Supplemental Table 5: Spontaneous Cor.4U cardiomyocyte electrophysiology values across 14 days of culture. BPM: beats per minute, FPD: field potential duration, stdev: standard deviation. Ordinary one-way ANOVA with multiple comparisons to day 7 using Holm-Sidak correction; significant p-values are bolded. Cohen effect size indicated (absolute value), relative to day 7 measurements.

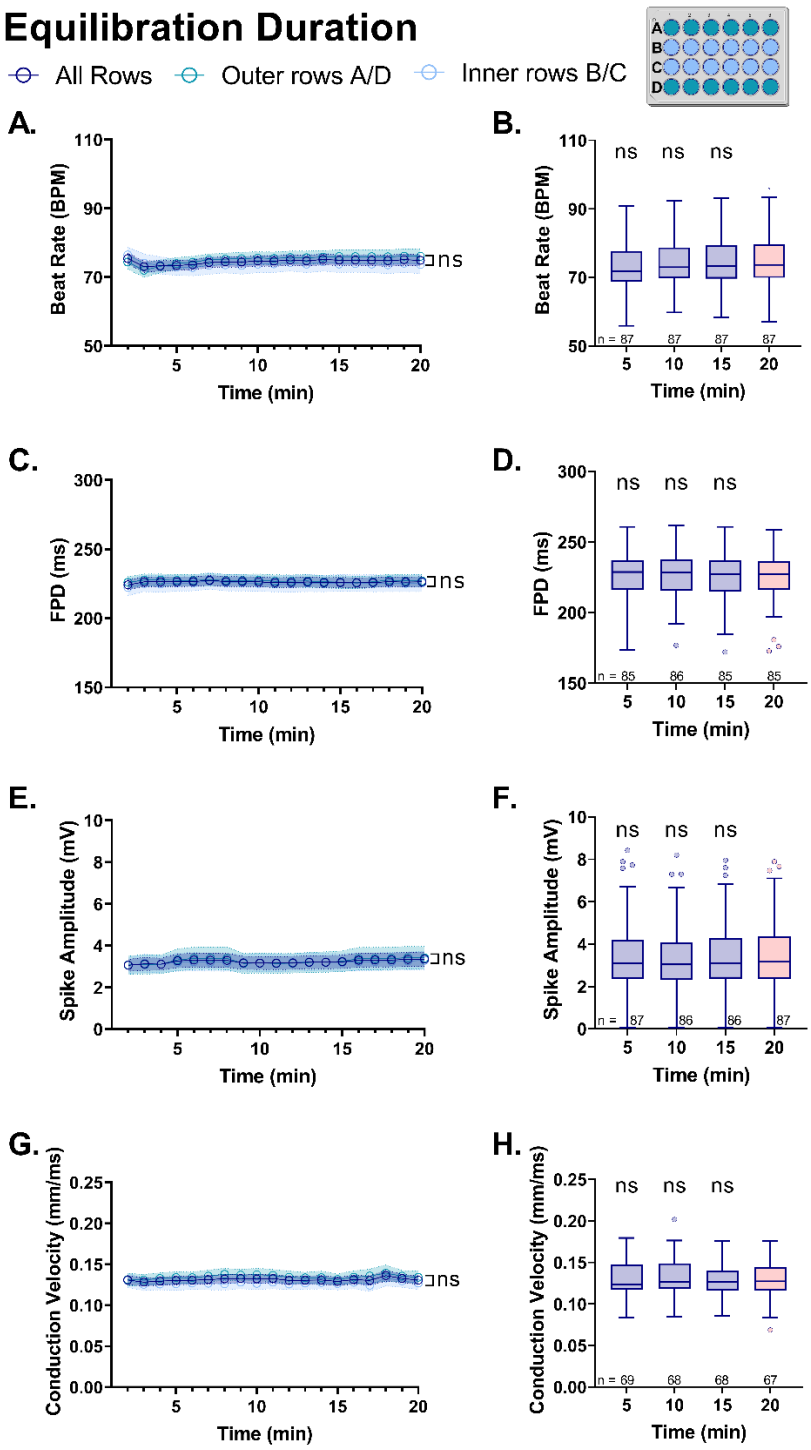
Day	Beat Rate (BPM)				FPD (ms)				Spike Amplitude (mV)				Conduction Velocity (mm/ms)			
	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value
2	73.01 ± 4.68	0.3025	0.289	123	182.0 ± 13.3	**<0.0001**	3.071	88	0.228 ± 0.153	**<0.0001**	2.723	144	0.072 ± 0.016	**<0.0001**	2.978	31
3	72.95 ± 5.51	0.3025	0.284	136	204.8 ± 14.7	**<0.0001**	1.548	86	0.528 ± 0.224	**<0.0001**	2.449	144	0.076 ± 0.011	**<0.0001**	3.051	92
4	78.39 ± 7.64	**0.0101**	0.453	119	207.5 ± 15.7	**<0.0001**	1.343	75	0.898 ± 0.379	**<0.0001**	2.092	120	0.086 ± 0.014	**<0.0001**	2.364	94
5	81.62 ± 9.31	**<0.0001**	0.782	143	212.1 ± 19.6	**<0.0001**	0.956	95	1.796 ± 0.752	**<0.0001**	1.235	139	0.105 ± 0.018	**<0.0001**	1.259	130
6	77.61 ± 9.05	0.0614	0.322	144	224.6 ± 23.6	0.1229	0.255	96	2.665 ± 1.120	**0.0046**	0.495	138	0.121 ± 0.018	**0.0432**	0.484	127
7	74.88 ± 7.87	-	-	87	229.9 ± 17.6	-	-	80	3.353 ± 1.616	-	-	87	0.131 ± 0.023	-	-	67
8	74.61 ± 9.09	0.9701	0.032	72	243.5 ± 21.0	**0.0001**	0.702	68	4.363 ± 2.326	**0.0007**	0.521	72	0.138 ± 0.028	0.1167	0.273	60
9	74.61 ± 7.43	0.9701	0.353	48	243.5 ± 20.0	**0.0004**	0.722	48	5.618 ± 2.270	**<0.0001**	1.150	48	0.151 ± 0.028	**0.0007**	0.781	44
10	73.55 ± 8.14	0.5882	0.166	88	248.2 ± 20.5	**<0.0001**	0.958	62	7.315 ± 3.406	**<0.0001**	1.486	88	0.181 ± 0.034	**<0.0001**	1.723	80
11	69.84 ± 7.44	**0.0026**	0.658	48	258.4 ± 21.5	**<0.0001**	1.451	48	6.918 ± 2.652	**<0.0001**	1.623	48	0.181 ± 0.038	**<0.0001**	1.592	44
12	67.70 ± 7.06	**<0.0001**	0.960	48	260.4 ± 21.2	**<0.0001**	1.565	48	7.321 ± 2.725	**<0.0001**	1.771	48	0.221 ± 0.058	**<0.0001**	2.040	44
13	66.12 ± 7.34	**<0.0001**	1.151	48	261.3 ± 23.4	**<0.0001**	1.517	48	7.467 ± 2.760	**<0.0001**	1.819	48	0.200 ± 0.048	**<0.0001**	1.833	46
14	71.55 ± 7.91	0.0950	0.422	48	236.5 ± 22.8	0.1229	0.324	47	8.093 ± 3.092	**<0.0001**	1.921	48	0.212 ± 0.039	**<0.0001**	2.530	45



Supplemental Figure 4. Daily changes in Cor.4U cardiomyocyte electrophysiology measurements (external pacing, 2 Hz). During two weeks of cell culture, **A)** field potential duration (FPD) remained relatively stable while **B)** depolarization spike amplitude and **C)** conduction velocity increased. Daily measurements are shown as a box-and-whisker plot (all values shown, min to max), with statistical comparison versus day 7 values (shown in red). ANOVA with Holm-Sidak correction for multiple comparisons. *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. Number of replicates indicated.

Supplemental Table 6: Paced (2 Hz) Cor.4U cardiomyocyte electrophysiology values across 14 days of culture. FPD: field potential duration, stdev: standard deviation. Ordinary one-way ANOVA with multiple comparisons to day 7 using Holm-Sidak correction; significant p-values are bolded. Cohen effect size indicated (absolute value), relative to day 7 measurements.

Day	FPD (ms)				Spike Amplitude (mV)				Conduction Velocity (mm/ms)			
	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value
3	190.0 ± 14.8	0.1748	0.775	20	0.719 ± 0.169	**<0.0001**	2.927	34	0.072 ± 0.013	**<0.0001**	3.092	30
4	191.5 ± 14.4	0.0804	0.686	34	1.040 ± 0.337	**<0.0001**	2.549	60	0.083 ± 0.015	**<0.0001**	2.356	42
5	192.5 ± 19.9	0.0912	0.522	44	1.677 ± 0.594	**<0.0001**	1.817	67	0.097 ± 0.016	**<0.0001**	1.553	50
6	198.8 ± 22.3	0.8715	0.160	49	2.668 ± 0.991	**0.0130**	0.790	68	0.114 ± 0.018	**0.0406**	0.614	55
7	201.9 ± 15.9	-		36	3.623 ± 1.393	-	-	48	0.126 ± 0.021	-	-	44
8	212.4 ± 14.5	0.0753	0.690	36	5.084 ± 2.058	**0.0023**	0.831	36	0.130 ± 0.019	0.4940	0.200	34
9	204.6 ± 13.7	0.8715	0.182	37	6.015 ± 2.338	**<0.0001**	1.243	36	0.147 ± 0.020	**0.0011**	1.024	33
10	201.9 ± 18.3	>0.9999	0.000	44	7.130 ± 2.818	**<0.0001**	1.578	48	0.157 ± 0.026	**<0.0001**	1.312	45
11	215.0 ± 14.7	**0.0121**	0.856	36	6.919 ± 2.863	**<0.0001**	1.464	39	0.161 ± 0.029	**<0.0001**	1.382	32
12	214.4 ± 13.5	**0.0168**	0.848	37	7.810 ± 2.593	**<0.0001**	2.012	37	0.175 ± 0.039	**<0.0001**	1.564	35
13	210.5 ± 17.5	0.1748	0.514	36	8.179 ± 2.897	**<0.0001**	2.004	36	0.189 ± 0.037	**<0.0001**	2.094	34
14	201.9 ± 16.7	>0.9999	0.000	38	8.383 ± 3.376	**<0.0001**	1.843	43	0.198 ± 0.042	**<0.0001**	2.168	38

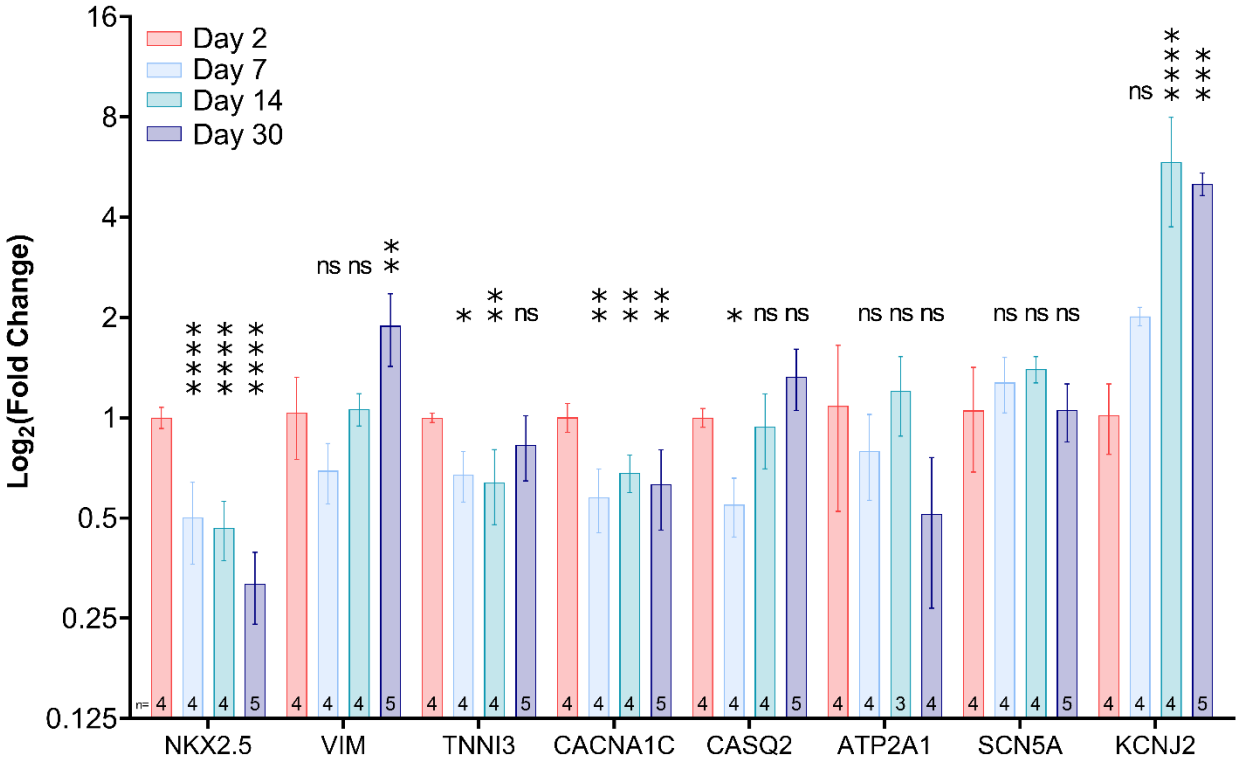


Supplemental Figure 5. Cor.4U cardiomyocyte equilibration duration. Cor.4U cardiomyocyte electrophysiology data was collected continuously for 20 minutes (7 days post-defrost) during spontaneous beating. **A,B)** Spontaneous beating rate. **C,D)** Field potential duration (FPD). **E,F)** Depolarization spike amplitude. **G,H)** Conduction velocity. **Left:** Row location measurements are reported as the average of all wells across the MEA plate (purple), outer rows (green), or inner rows (blue). Open circle = mean, shaded region = 95% confidence interval. Statistical comparison was performed versus the 20-minute timepoint (ANOVA with Holm-Sidak correction for multiple comparisons). All Rows: n=67-88; Outer rows A/D: n=34-44; inner rows B/C: n=32-44. **Right:** Equilibration time measurements are for all wells shown as a Tukey box-and-whisker plot, with outliers indicated. Statistical comparisons at five-minute intervals (purple box plots) versus the end of equilibration at the 20-minute timepoint (red box plot). ANOVA with Holm-Sidak correction for multiple comparisons. Number of replicates is indicated for each time point.

Supplemental Table 7: Spontaneous Cor.4U cardiomyocyte electrophysiology values across 20 minutes of equilibration in a microelectrode array system. BPM: beats per minute, FPD: field potential duration, stdev: standard deviation. Ordinary one-way ANOVA with multiple comparisons to minute 20 using Holm-Sidak correction; significant p-values are bolded. Cohen effect size indicated (absolute value), relative to 20 minute measurements.

Minute	Beat Rate (BPM)				FPD (ms)				Spike Amplitude (mV)				Conduction Velocity (mm/ms)			
	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value	mean ± stdev	p value	|effect size|	n value
2	75.40 ± 7.19	>0.9999	0.069	87	224.1 ± 16.8	0.9998	0.128	86	3.063 ± 1.339	0.9857	0.195	84	0.131 ± 0.024	>0.9999	0.000	70
3	73.07 ± 7.78	0.8631	0.231	86	226.4 ± 16.6	>0.9999	0.006	86	3.127 ± 1.448	0.9983	0.147	85	0.128 ± 0.021	>0.9999	0.136	69
4	73.28 ± 7.21	0.9372	0.212	87	226.3 ± 16.9	>0.9999	0.000	85	3.110 ± 1.341	0.9973	0.164	84	0.129 ± 0.020	>0.9999	0.093	69
5	73.42 ± 6.82	0.9651	0.198	87	226.4 ± 16.4	>0.9999	0.006	85	3.268 ± 1.567	>0.9999	0.053	87	0.130 ± 0.022	>0.9999	0.044	69
6	73.59 ± 6.87	0.9862	0.175	87	226.5 ± 16.5	>0.9999	0.012	86	3.309 ± 1.603	>0.9999	0.027	88	0.131 ± 0.021	>0.9999	0.000	69
7	74.25 ± 7.06	>0.9999	0.084	87	227.4 ± 17.2	>0.9999	0.063	85	3.305 ± 1.606	>0.9999	0.030	88	0.131 ± 0.022	>0.9999	0.000	68
8	74.42 ± 6.79	>0.9999	0.063	87	226.5 ± 16.8	>0.9999	0.012	86	3.294 ± 1.616	>0.9999	0.037	88	0.132 ± 0.022	>0.9999	0.044	69
9	74.40 ± 7.24	>0.9999	0.063	87	226.5 ± 17.0	>0.9999	0.012	86	3.173 ± 1.453	0.9996	0.117	86	0.133 ± 0.022	>0.9999	0.089	70
10	74.68 ± 7.09	>0.9999	0.027	87	226.3 ± 16.5	>0.9999	0.000	86	3.169 ± 1.456	0.9996	0.120	86	0.132 ± 0.023	>0.9999	0.043	68
11	74.65 ± 7.24	>0.9999	0.030	87	225.8 ± 17.0	>0.9999	0.029	87	3.163 ± 1.459	0.9996	0.123	86	0.133 ± 0.022	>0.9999	0.089	70
12	74.99 ± 7.24	>0.9999	0.015	87	225.9 ± 17.1	>0.9999	0.023	85	3.175 ± 1.441	0.9996	0.116	86	0.130 ± 0.021	>0.9999	0.045	70
13	74.73 ± 7.30	>0.9999	0.020	86	226.2 ± 17.3	>0.9999	0.006	85	3.206 ± 1.469	0.9997	0.095	86	0.131 ± 0.021	>0.9999	0.000	69
14	75.14 ± 7.49	>0.9999	0.034	87	225.8 ± 16.9	>0.9999	0.029	86	3.207 ± 1.477	0.9997	0.094	86	0.131 ± 0.023	>0.9999	0.000	69
15	74.93 ± 7.41	>0.9999	0.007	87	225.8 ± 17.3	>0.9999	0.029	85	3.224 ± 1.489	0.9997	0.083	86	0.129 ± 0.021	>0.9999	0.091	68
16	74.97 ± 7.45	>0.9999	0.012	87	225.6 ± 17.1	>0.9999	0.040	86	3.303 ± 1.581	>0.9999	0.031	87	0.131 ± 0.023	>0.9999	0.000	71
17	74.91 ± 7.53	>0.9999	0.004	87	225.9 ± 17.1	>0.9999	0.023	85	3.306 ± 1.575	>0.9999	0.029	87	0.130 ± 0.024	>0.9999	0.043	69
18	74.77 ± 7.73	>0.9999	0.014	87	226.4 ± 16.7	>0.9999	0.006	86	3.315 ± 1.587	>0.9999	0.024	87	0.136 ± 0.029	0.9353	0.191	72
19	75.09 ± 7.91	>0.9999	0.027	87	226.1 ± 16.8	>0.9999	0.012	86	3.340 ± 1.597	>0.9999	0.008	87	0.133 ± 0.021	>0.9999	0.091	67
20	74.88 ± 7.87	-	-	87	226.3 ± 17.5	-	-	85	3.353 ± 1.616	-	-	87	0.131 ± 0.023	-	-	67

Changes in Gene Expression



Supplemental Figure 6. Changes in Cor.4U gene expression across one month of culture. Genes of interest included homeobox protein (*NKX2.5*; a marker of heart development), vimentin (*VIM*; a fibroblast marker), troponin I (*TNNI3*; a marker of cardiac muscle contraction), voltage-dependent L-type calcium channel alpha-1C (*CACNA1C*, a marker of calcium influx), calsequestrin-2 (*CASQ2*, a marker of calcium storage), sarcoplasmic/endoplasmic reticulum calcium ATPase 1 (*ATP2A1*, a marker of calcium reuptake), sodium channel protein type 5 subunit alpha (*SCN5A*, a marker of action potential propagation), and inward rectifier potassium channel 2 (*KCNJ2*, a marker of the resting membrane potential). Ordinary one-way ANOVA with multiple comparisons to day 2 (shown in red) using Holm-Sidak correction; *p<0.05, **p<0.01, ***p<0.001, ****p<0.0001. Number of replicates indicated.

Supplemental Table 8: Example reporting datasheet. Key experimental details in human induced pluripotent stem cell-derived cardiomyocyte (hiPSC-CM) microelectrode array (MEA) studies.

hiPSC-CM Parameters	
Manufacturer	
Cell line	
Product number	
Donor sex	
Donor age	
Cardiomyocyte purity (%)	
Lot number(s)	
Number of vials used	
Plating density (cells/well)	
Experimental Parameters	
Number of MEA plates used	
Time post-defrost (days)	
Equilibration duration (min)	
Temperature (°C)	
CO_2 (%)	
Recording length (s)	
Sampling frequency (kHz)	
Pacing Settings	
Frequency (Hz)	
Stimulus duration (μs)	
Voltage (mV)	
Current (μA)	
Stimulus type (monophasic/biphasic/etc)	
Analysis Parameters	
Number of electrodes analyzed	
Length of time signals were averaged over (s)	